

Forough Mahpouya (she/her)

Ph.D. Candidate in Industrial and Systems Engineering

Department of Industrial and Systems Engineering, University at Buffalo, The State University of New York

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RESEARCH AREAS

Causal inference and representation learning; sequential experimentation and ranking; personalized treatment selection and adaptive care; aging and patient–caregiver systems.

EDUCATION

Ph.D. Candidate, Industrial and Systems Engineering 2022–present

University at Buffalo, The State University of New York, Buffalo, NY

Expected: June 2027

Dissertation: Quantitative Modeling of Health Outcomes in Patient–Caregiver Dyads in End-of-Life Care

Advisors: Alexander Nikolaev and Sabrina Casucci

M.Sc., Industrial Engineering 2017–2019

K. N. Toosi University of Technology, Tehran, Iran

Thesis: A Bootstrap Approach to Introduce a Robust Multivariate Control Chart

Advisors: Hamid Shahriari and Emad Roghanian

B.Sc., Industrial Engineering 2006–2010

Iran University of Science and Technology, Tehran, Iran

SELECTED RESEARCH EXPERIENCE

Doctoral Researcher (D) 2023–present

Department of Industrial and Systems Engineering, University at Buffalo, Buffalo, NY

1. **Balancing representations for continuous interventions and time-to-event outcomes.** Led theoretical development of a neural balancing framework for continuous-dose interventions with survival outcomes; derived identification conditions and connected the learning objective to hazard modeling. Authored a technical manuscript with proofs and simulation design, and prepared an implementation plan and real-data analysis protocol.
2. **Caregiver burden–benefit measurement and outcome prediction.** Developed and validated the Caregiver Burden–Benefit Index (CareBI) using NHATS/NSOC data; constructed predictive models of adverse caregiver outcomes and examined predicted probabilities across caregiver profiles and service-support groups.
3. **Heterogeneous treatment effects and subgroup identification.** Authored a structured review manuscript classifying approaches to subgroup identification and treatment-effect estimation, including model-based partitioning, uplift and loss-based learners, causal trees and forests, and representation learning. Developed a decision map linking data structure and clinical objectives to method selection, with guidance on stability, interpretability, and fairness.
4. **Longitudinal analysis of diabetes and cognitive decline.** Analyzed Health and Retirement Study (HRS) data to compare cognitive decline among older adults with and without diabetes; aligned diabetes status and participant characteristics with the period preceding cognitive impairment and applied adversarial balancing to account for measured differences between groups.
5. **Fairness in glomerular filtration rate prediction.** Reviewed glomerular filtration rate (GFR) measurement and fairness literature and assessed the potential application of adversarial balancing-based representation learning to GFR prediction; co-authored an NIDDK data-request proposal specifying the study rationale, methodological approach, and required biomarkers.

Researcher, Anomaly Detection in Inkjet Printing

2022–2023

Department of Industrial and Systems Engineering, University at Buffalo, Buffalo, NY

Led theory development and preliminary methodological design for classification-based anomaly detection in inkjet printing through analysis of droplet pinch-off behavior; documented findings, identified methodological gaps, and proposed improvements.

Research Collaborator

2017–2022

Department of Industrial Engineering, K. N. Toosi University of Technology, Tehran, Iran

Authored a structured literature report classifying mean-vector estimators for multivariate populations by number of samples, sample size, and the presence of outliers; organized a literature-review manuscript on applications of degradation-data models to product quality improvement and stability in the food industry.

PUBLICATIONS AND MANUSCRIPTS

Refereed Journal Articles (J)

1. **Mahpouya, F.**, Burris, C. J., Paul, H., & Nikolaev, A. (2026). Maximizing the expected value of experimentation for finding top- κ rank via aggregation of pairwise comparisons. *IIE Transactions*, 1–20.
2. **Mahpouya, F.**, Shahriari, H., & Roghanian, E. (2022). Design of a robust T^2 control chart, a re-sampling approach. *Quality and Reliability Engineering International*, 38(2), 924–940.

Manuscripts Under Review or Revision (R)

1. Das, S., Casucci, S., **Mahpouya, F.**, & Nikolaev, A. (2026). Treatment effect estimation from observational data in healthcare: Study designs, inherent biases, and a propensity score-focused quality assessment scale. Manuscript under revision for *Health Services and Outcomes Research Methodology*.
2. **Mahpouya, F.**, Das, S., Nikolaev, A., & Casucci, S. (2026). Use of observational causal inference in dementia management studies: A systematic literature review. Manuscript under revision for *PLOS Digital Health*.
3. Walker, R. J., Linde, S., Egede, L. E., **Mahpouya, F.**, Nikolaev, A., Fraser, R., & Martinez, J. (2026). Assessing the Healthy Eating Index and health: Psychometric properties and prediction of cardiometabolic outcomes. Manuscript under review at *The Journal of Nutrition*.

Works in Progress (W)

1. **Mahpouya, F.**, Nikolaev, A., & Casucci, S. (2026). Adversarial balancing for estimating the causal effect of caregiver burden on patient long-term-care transitions. Prepared for submission.
2. **Mahpouya, F.**, Casucci, S., & Sullivan, S. (2026). A population-based, bidirectional measure of caregiver risk and resilience: The Caregiver Burden–Benefit Index (CareBI). Prepared for submission.
3. Rogers, A. H., Fligelman, S. J., Pester, B. D., Pielech, M., Klages, K. L., **Mahpouya, F.**, Nikolaev, A., Groenewald, C. B., Ketterl, T. G., Bevers, L. M., Dietel, K. E., & Palermo, T. M. (2026). Pain management and opioid use in adolescents and young adults with cancer: A qualitative analysis of patient and caregiver experiences. Prepared for submission.
4. Rogers, A. H., Fligelman, S. J., Pester, B. D., Pielech, M., Klages, K. L., **Mahpouya, F.**, Nikolaev, A., Groenewald, C. B., Ketterl, T. G., Bevers, L. M., Dietel, K. E., & Palermo, T. M. (2026). Pain management and opioid use in adolescents and young adults with cancer: A quantitative analysis using Ecological Momentary Assessment (EMA). Prepared for submission.
5. Molazadeh, M., Casucci, S., **Mahpouya, F.**, Nikolaev, A., & Das, P. S. (2026). Hospice Enrollment at the End of Life: Associations Between Long-Term Care Status, Functional Dependence, and Comorbidities. Prepared for submission.

CONFERENCE PRESENTATIONS (C)

1. **Mahpouya, F.**, Nikolaev, A., & Casucci, S. (2026). Adversarial balancing for estimating the causal effect of caregiver burden on patient long-term care transitions. *IISE Annual Conference & Expo*, Arlington, TX.
2. **Mahpouya, F.**, Casucci, S., Sullivan, S., & Barrick, C. (2025). Representing caregiver burden in observational studies: Development of the Caregiver Burden Index (CareBI) using NSOC. *INFORMS Annual Meeting*, Atlanta, GA.
3. **Mahpouya, F.**, Shahriari, H., & Roghanian, E. (2020). Use of resampling techniques to design a robust Hotelling's T^2 control chart for limited data. *16th Iran International Industrial Engineering Conference*, Alzahra University, Tehran, Iran. Conference record
4. **Mahpouya, F.**, & Shahriari, H. (2019). A bootstrapping approach to introduce a robust multivariate control chart. *15th Iran International Industrial Engineering Conference*, Yazd University, Yazd, Iran. Conference record

TEACHING EXPERIENCE

Teaching Assistant and Guest Lecturer 2022–present
University at Buffalo, Buffalo, NY

Course areas: simulation and stochastic models; production planning and control; applied probability and statistics; healthcare systems engineering; computer-integrated manufacturing; social-network behavior models.

IE 505: Production Planning and Control Spring 2026

IE 477: Simulation Modeling and Analysis Fall 2023, 2024, 2025, 2026

- Led 12 recitation sessions and two course lectures per semester in Fall 2023–2025; designed six weeks of recitation material using @RISK and Arena.

IE 406/506: Computer-Integrated Manufacturing Fall 2023, 2025

- Led four lab tutorials each semester using Autodesk Fusion 360 for 3D CAD modeling and manufacturing.

IE 551: Simulation and Stochastic Models Spring 2024, 2025

- Delivered a course lecture and held group-specific active office hours.

IE 511: Social Network Behavior Models Spring 2024, 2025

IE 585: Industrial Engineering Principles and Practice in Healthcare Fall 2024

- Delivered a guest lecture on PICO question formulation and analysis of complex observational healthcare data for interpretable evidence.

EAS 305: Applied Probability and Statistics Fall 2022–Summer 2023

- Led 54 recitations in Fall 2022 and 30 in Spring 2023; designed ten weeks of recitation material; delivered a Minitab lab and an exam-review lecture each semester.

INDUSTRY AND PROFESSIONAL EXPERIENCE

Industrial Engineer / Systems Optimizer 2012–2016
Arioplast, photo-frame profile manufacturing, Rasht, Iran

- Reorganized operations and workflows, freeing two employees to staff two newly added production lines.
- Co-designed an automated weighing and pricing system that categorized products, monitored manufacturing waste in real time, and enabled immediate pricing of customer orders.
- Developed process instructions and operating manuals and trained staff on the new procedures.

Part-Time Technical Mentor 2010–2012; 2015–2017
Private training institutes, Rasht and Sari, Iran

- Designed and delivered tutorials and mentored students in ICDL, Visio, and MATLAB.

HONORS AND AWARDS

- **University at Buffalo ISE Graduate Student Teaching Award**, 2025–2026.
- **IISE Future Faculty Fellows (3F) Program**, selected fellow, 2025–2026.
- **Exceptionally Talented Student designation**, M.Sc. program, K. N. Toosi University of Technology, 2021.
- Ranked **3rd of 36** students in the M.Sc. program, K. N. Toosi University of Technology, 2019.
- Ranked in the **top 1% of 15,800 participants** in Iran's national M.Sc. entrance examination in Industrial Engineering, 2017.

PROFESSIONAL AND ACADEMIC SERVICE

- Reviewer for the *Journal of Nursing Measurement*.
- Contributed to the University at Buffalo's *Effective Pedagogy for Teaching Assistants* program by sharing approaches to recitation design and addressing teaching-assistant challenges through Universal Design for Learning, Spring 2025.
- Mentored undergraduate students on university projects and statistical analysis using MATLAB, 2010–2022.

METHODS AND TECHNICAL EXPERTISE

Causal and Machine Learning Methods: Causal inference; balancing and adversarial representation learning; continuous-treatment effect estimation; treatment-effect heterogeneity and subgroup analysis; longitudinal data analysis; survival and hazard modeling; neural networks; fairness-aware learning.

Statistical and Systems Methods: Scale development and psychometric validation; bifactor modeling; simulation and stochastic modeling; sequential experimentation and ranking; experimental design; anomaly detection; robust multivariate statistics and process control.

Applications: Healthcare outcome modeling; aging and patient–caregiver systems; cardiometabolic health; manufacturing quality and process improvement.

Programming and Research Tools: Python; R; MATLAB.

Technical Software: Arena; @RISK; Autodesk Fusion 360; Minitab.